Group 1 Execution Plan

PPO $\times$ Macro Adaptive Strategy

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Outline

1. Strategy Recap
2. Execution Design
3. Transaction Costs
4. Order Execution
5. Trade Lifecycle & Safeguards
6. Backtest $\rightarrow$ Live

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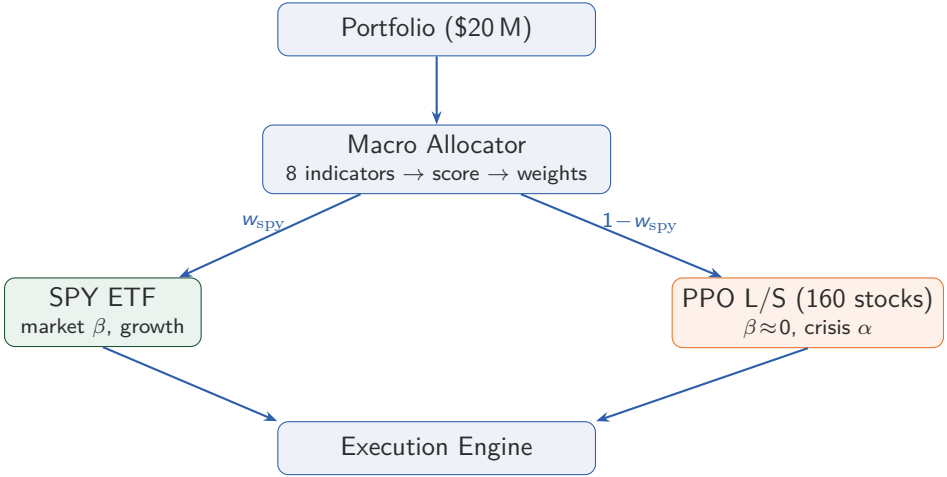
The Core Insight

- A **PPO reinforcement-learning agent** dynamically tunes 15 factor weights to build a **beta-neutral long/short equity portfolio** ($\beta \approx 0$, 80 L / 80 S).
- Beta neutrality unlocks significant **crisis alpha**—the portfolio profits when markets sell off.
- **Trade-off:** in bull markets, zero beta means limited upside.
- **Solution:** combine PPO with a **SPY position** ($\beta \approx 1$) and allocate capital dynamically:
  - Bull regime $\rightarrow$ more SPY weight $\rightarrow$ ride market growth
  - Crisis regime $\rightarrow$ more PPO weight $\rightarrow$ harvest crisis alpha
- The **macro allocator** reads 8 indicators, scores the regime, and shifts capital—fast into defence, slow back to offence.

Two-Layer Architecture



Dynamic Macro Allocation

- Score aggregates SPY/SMA distance, momentum, drawdown, VIX, yield curve spread via weighted vote.
- Crisis intensity ramps **fast** (EMA $\alpha = 0.70$) and decays **slowly** ($\alpha = 0.15$)—asymmetric by design.
- $w_{\mathrm{spy}} = w_{\mathrm{base}}(s) \times (1 - c); \qquad g_{\mathrm{ppo}} = 0.95\,(1 - w_{\mathrm{spy}})$.
- Sensitivity trades off *false-signal cost* (too sensitive $\rightarrow$ excess turnover) vs. *opportunity cost* (too slow $\rightarrow$ miss the regime shift).

Ramp-Up Period: Training the PPO Agent

- PPO begins with a **random policy**—early trades are exploratory and loss-making.
- Without a backstop, large random L/S positions would trigger **margin calls**, killing execution and crippling learning.
- **Our design:** for the first 2 years, fix **80 % to SPY** / **20 % gross to PPO**.
- SPY growth keeps the portfolio solvent; PPO explores within a small sandbox.
- After ramp-up, macro allocator takes over with full dynamic allocation.

	Ramp	**Post**
SPY wt	80%	dynamic
PPO gross	20%	dynamic
Duration	2 yr	ongoing
PPO policy	random	trained

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Execution Objectives

Lecture Goal	Our Implementation
Trade at identical times	Daily rebalance 15 min after open via `Schedule.On`; signal & execution in the same event.
Trade at backtest prices or better	Three modes: passthrough / limit-at-mid / MOC; limit has 90 s timeout $\rightarrow$ market fallback.
Minimise slippage	0.5 bps one-way model; limit targets mid-price; less-liquid assets first.
Minimise commission	IB tiered: \$0.0035/sh, \$0.35 min; identical model in backtest & live.
Minimise tax	30 % dividend WHT tracked per event; penalised in PPO reward function.

Timing & Instrument Selection

When do we trade?

- Signals generated **15 min after open** daily (previous close already known).
- Execution follows immediately within the same scheduled event.
- No after-hours trading—wider spreads and lower liquidity make it unsuitable.
- **MOC mode** available for close-price fidelity (orders before 3:45 pm).

What do we trade?

Layer	Instr.	Rationale
Macro	SPY ETF	Liquid, no roll
PPO	160 US eq.	Factor-level tilt

- SPY over ES futures: no roll risk, simpler bookkeeping.
- Dividend tax mitigated by WHT-aware reward.
- Universe: top 400 by \$ vol, price $>$ \$5, with fundamentals.

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Slippage & Commission Models

Cost	Model	Parameters & Rationale
Slippage	`SpreadSlippageModel`	0.5 bps one-way (1 bps round-trip). Conservative for liquid US large-caps; models half the bid-ask crossing cost.
Commission	`IBTieredFeeModel`	\$0.0035/sh, \$0.35 min, cap 0.5 % of trade value. Matches IB tiered tier 1.
Borrow	Flat annual rate	3 % p.a. live / 0 % backtest. Daily charge: $\lvert\text{short val}\rvert \times r/252$.

Applied to **every equity** via `OnSecuritiesChanged`. Identical models in backtest and live minimise the paper-to-real gap. The goal is to execute at the **mid-price**; VWAP is a viable alternative if order sizes grow.

Dividend Withholding Tax

- Non-US residents face **30 % US withholding tax** on equity dividends—a material drag for a 160-stock portfolio.
- `DividendTaxModel` scans every data slice for dividend events on long holdings:

$$\text{WHT} = \text{Distribution}_{\text{per share}} \times \text{Qty} \times 30\,\%$$

- Tax is injected into the reward function ($-\lambda_{\text{tax}} \cdot \text{tax}\,\%$), so PPO *learns* to penalise high-dividend longs.
- Full audit trail: symbol, qty, gross dividend, tax, cumulative total—persisted to `ObjectStore`.
- Short positions pay the dividend to lender (broker-handled); no WHT applies.

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Three Execution Modes

Mode	Mechanism	When to Use	Slippage
Passthrough	`SetHoldings`	Backtest; low-urgency	QC default fill
Limit	Limit at mid, 90 s timeout $\rightarrow$ market	**Default live**	Targets mid
MOC	`MarketOnCloseOrder`	Close-price match	Closing auction

- Passthrough ensures **zero code divergence** between backtest and live.
- Limit mode adds **price improvement** via mid-price targeting; falls back safely to market on timeout.
- MOC mode ensures close-price execution if signals are generated from close prices (submitted before 3:45 pm cutoff).

Smart Order Routing

- **Less-liquid-first ordering**: orders are sorted by *descending* bid-ask spread so the hardest fills go first via limit; easier fills follow with market if needed.
- **Chunked execution** for large orders ($> \$500\,\mathrm{K}$): split into 4 child orders to reduce market impact and average out timing uncertainty.
- **Timeout fallback**: unfilled limits after 90 s are cancelled and resubmitted as market orders, guaranteeing the full rebalance completes each day.
- **Latency**: daily rebalance makes sub-ms latency unnecessary. Standard API round-trip ($\sim$50–200 ms) is sufficient. Co-location cost unjustified at our frequency.
- **Maker vs. Taker**: limit orders earn maker rebates on some venues; market-order fallback pays taker fees. Net saving is modest but positive over many fills.

Short Selling & Borrow Costs

- PPO holds ∼80 short positions at any time (∼47.5 % of PPO capital).
- **Challenges**:
  - High borrow demand during crises → inventory shortages, rate spikes
  - Wider spreads on hard-to-borrow names
- **Our mitigations:**
  - Liquid universe (top-400 by $ vol) → mostly "general collateral"
  - 3 % p.a. flat borrow in live
  - `RiskGuard`: per-order spread & borrow warnings
  - Daily short-inventory summary logged

Recall: RSX Borrow Spike

Borrow fee jumped from ∼3 % to >31 % around Russia's invasion of Ukraine (Feb 24, 2022).

Our safeguard: liquid-universe filter avoids special-situation names. `RiskGuard` alerts if any spread > 200 bps.

If borrow becomes unavailable, `check_short_feasibility` flags the symbol before order submission.

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Trade Signal Types

Signal	Definition	Our Trigger
Entry	Open a new L or S position	PPO factor weights $\rightarrow$ scores $\rightarrow$ top/bottom 80
Exit	Close all positions	Crisis override ($g_{\text{ppo}} \rightarrow 0$); stock leaves universe
Rebalance	Adjust position size	Daily macro re-weight; beta adjustment; vol targeting; portfolio investment/redemption

Each signal creates, updates, or closes a **Trade Order** tracked by `TradeTracker` with unique IDs. A rebalance signal closes the existing order at current price and opens a new one—preserving the full P&L audit trail.

Five-Layer Record Keeping

#	Layer	Purpose
1	Signal + snapshot	Data & market state at signal time; compare live vs. backtest signals.
2	Batch signals	All 160 target values + macro metadata per rebalance.
3	Fill execution	Actual fill price, qty, commission; slippage vs. mid computed per fill.
4	Daily MTM	Unrealised P&L per holding at close; for daily return, vol, drawdown series.
5	Realised P&L	`TradeTracker`: profit per trade ID; includes dividends & commissions.

All five layers run in **both backtest and live** and persist to QC `ObjectStore` in JSON/JSONL. This enables full post-trade analysis and broker reconciliation.

Risk Guards & Reconciliation

Pre-Trade Risk Guards

- **Circuit breaker:** halt if daily loss $> 5\,\%$.
- **Gross leverage cap:** block rebalance if $> 1.5\times$.
- **Single position:** reject if any holding $> 20\,\%$.
- **Spread alert:** warn if bid-ask > 200 bps (non-blocking).
- E-mail notification on circuit-breaker trigger.

End-of-Day Reconciliation

- Broker vs. target: flag deviation $> 2\,\%$ per symbol.
- Cross-check broker vs. `TradeTracker` open positions (qty mismatch, unexpected holdings).
- Dividend income synced into `TradeTracker` (live mode).
- Auto-correction available but disabled by default.
- Execution stats logged: avg/max slippage, commissions.

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Model Registry & Seamless Transition

- **Model Registry** saves full PPO state to `ObjectStore`: policy weights, value weights, *and both optimizer states*.
- Backtest saves a checkpoint every 20 days plus a final `backtest_final` checkpoint with Sharpe, cumulative P&L, and max drawdown.
- **Live** loads the best checkpoint by Sharpe and *continues training identically*—optimizer momentum and learning-rate state are fully preserved.
- Fallback chain: Registry $\rightarrow$ ObjectStore (legacy) $\rightarrow$ filesystem $\rightarrow$ random weights (with alert).

Design Principle

Backtest *is* the first phase of live trading. Every cost model, risk guard, and record layer is active in **both** modes—zero code divergence.

Execution Plan Summary

Dimension	Decision
Automated or manual?	Fully automated (QC $\rightarrow$ IB paper/live)
Data source	QuantConnect equity + CBOE VIX + FRED yield curve
Signal generation	15 min after market open (daily)
Execution time	Immediately after signal; MOC available
SPY instrument	SPY ETF (no rolling, simpler than ES futures)
PPO instruments	Top-400 liquid US equities; long 80 / short 80
Order type	Limit at mid, 90 s timeout $\rightarrow$ market
Slippage model	0.5 bps one-way (`SpreadSlippageModel`)
Commission	IB tiered \$0.0035/sh (`IBTieredFeeModel`)
Borrow cost	3 % p.a. (live); daily accrual on shorts
Dividend tax	30 % WHT on long dividends
Risk controls	Circuit breaker -5 %; leverage $\leq 1.5\times$; pos ≤ 20 %
Record keeping	5-layer (signal / batch / fill / MTM / realised)

Thank You

Questions & Discussion

Appendix A1: PPO Network Architecture

Policy Network $\pi_\theta(a \mid s)$: 3-layer MLP (128 hidden, Tanh), outputs mean $\mu(s)$ and learned log-std $\log \sigma$.

$$a \sim \mathcal{N}\big(\mu_\theta(s),\ \mathrm{diag}(\sigma^2)\big), \qquad w = \frac{\tanh(a)}{\|\tanh(a)\|_1}$$

$w \in \mathbb{R}^{15}$ are L1-normalised factor weights.

Value Network $V_\phi(s)$: 3-layer MLP (128 hidden, Tanh) $\rightarrow$ scalar value estimate.

State ($d = 87$): 12 core features + 5×15 factor statistics.
Core: log(equity), cash%, gross lev, β, w_L, w_S, n_{pos}, turnover, tc%, borrow%, β_{last}, TE.
Per factor ($k = 1 \ldots 15$): $\overline{\text{IC}}_k$, $\sigma_{\text{IC},k}$, $t_{\text{IC},k}$, $\overline{\text{QS}}_k$, $\sigma_{\text{QS},k}$.

Action ($d = 15$): one weight per alpha factor.

Appendix A2: PPO Objective & GAE

Clipped Surrogate Objective:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\big(r_t(\theta)\,\hat{A}_t,\ \ \text{clip}(r_t(\theta),\,1-\epsilon,\,1+\epsilon)\,\hat{A}_t\big)\Big], \quad r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)},\ \epsilon = 0.2$$

Generalised Advantage Estimation ($\gamma = 0.99,\ \lambda = 0.95$):

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l\,\delta_{t+l}, \qquad \delta_t = r_t + \gamma\,V(s_{t+1}) - V(s_t)$$

Value loss: $L^V(\phi) = \frac{1}{N}\sum_t (V_\phi(s_t) - \hat{R}_t)^2, \quad \hat{R}_t = \hat{A}_t + V(s_t).$

Training: 10 epochs, batch 256, gradient clip 1.0. Updated every 5 trading days (after ramp-up).

Appendix A3: Macro Score — 8 Indicators

$$s_{\text{macro}} = \frac{\sum_i v_i \cdot \omega_i}{\sum_i \omega_i}, \qquad v_i \in \{-1,\ 0,\ +1\}$$

i	**Indicator**	ω_i	**Bullish** $(+1)$	**Bearish** (-1)
1	SPY / SMA200 − 1	2.0	$> 3\%$	$< -2\%$
2	SMA50 / SMA200 − 1	1.5	$> 1\%$	$< -1\%$
3	252-d momentum	1.0	$> 5\%$	$< -5\%$
4	Drawdown from 252-d hi	2.0	$> -3\%$	$< -8\%$
5	VIX level	2.5	< 16	> 22
6	Yield 10Y−2Y	1.0	> 0.50	< 0
7	20-d momentum	2.0	$> 3\%$	$< -4\%$
8	50-d momentum	1.5	$> 4\%$	$< -5\%$

Appendix A4: Crisis Intensity Computation

Four crisis signals (all clamped to $[0, 1]$):

$$\sigma_{\text{trend}} = \text{clamp}\left(\frac{-\text{trend} - 0.03}{0.22},\ 0,\ 1\right), \qquad \sigma_{\text{vol}} = \text{clamp}\left(\frac{\text{vol} - 0.05}{0.20},\ 0,\ 1\right),$$

$$\sigma_{\text{accel}} = \text{clamp}\left(\frac{\text{vol_accel} - 0.01}{0.09},\ 0,\ 1\right), \qquad \sigma_{\text{level}} = \text{clamp}\left(\frac{-s_{\text{macro}} - 0.10}{0.30},\ 0,\ 1\right).$$

Composite raw crisis:

$$c_{\text{raw}} = 0.50\ \min(\sigma_{\text{trend}},\ \sigma_{\text{vol}}) + 0.25\ \sigma_{\text{vol}} \cdot \sigma_{\text{accel}} + 0.25\ \sigma_{\text{level}}$$

Asymmetric EMA smoothing:

$$c_t = \alpha\, c_{\text{raw}} + (1 - \alpha)\, c_{t-1}, \qquad \alpha = \begin{cases} 0.70 & c_{\text{raw}} > c_{t-1} \text{ (fast ramp)} \\ 0.60 & \text{trend} > 0.05 \ \wedge\ s > -0.5 \\ 0.40 & \text{trend} > 0 \\ 0.15 & \text{otherwise (slow decay)} \end{cases}$$

Appendix A5: Dynamic Weight Allocation

Step 1 — Base SPY weight (C-style, since static weight $= 0$):

$$w_{\text{base}} = \text{clamp}\left(0.7 + \frac{s_{\text{macro}}}{\hat{\sigma}_{\text{long}}},\ -0.5,\ 1.0\right), \qquad \hat{\sigma}_{\text{long}} = \text{std}\left(s_{\text{buf}}[-260:]\right)$$

Step 2 — Crisis-adjusted SPY target:

$$w_{\text{spy}} = w_{\text{base}} \times (1 - c_t), \qquad w_{\text{spy}} \ \leq\ 1 - \frac{g_{\min}}{0.95} \ =\ 1 - \frac{0.30}{0.95} \approx 0.684$$

Step 3 — PPO gross exposure:

$$g_{\text{ppo}} = 0.95\,(1 - w_{\text{spy}})$$

Step 4 — PPO capital:

$$K_{\text{ppo}} = (1 - w_{\text{spy}}) \times \text{Equity}$$

Example: $s_{\text{macro}} = 0.4,\ c_t = 0.1,\ \hat{\sigma} = 0.25 \Rightarrow w_{\text{base}} = \text{clamp}(0.7 + 1.6,\ -0.5,\ 1.0) = 1.0$
$\Rightarrow w_{\text{spy}} = 1.0 \times 0.9 = 0.90\ \rightarrow$ capped at $0.684 \Rightarrow g_{\text{ppo}} = 0.95 \times 0.316 = 0.30$.

Appendix A6: 15 Alpha Factors

k	Factor	Formula (sign = desired direction)
1	mom_5	$P_t/P_{t-5} - 1$
2	mom_20	$P_t/P_{t-20} - 1$
3	mom_60	$P_t/P_{t-60} - 1$
4	rev_2	$-(P_t/P_{t-2} - 1)$
5	rev_5	$-(P_t/P_{t-5} - 1)$
6	vol_10	$-\mathrm{std}(r_{t-9:t})$
7	vol_20	$-\mathrm{std}(r_{t-19:t})$
8	hlv_20	$-\mathrm{mean}\big((H-L)/P\big)_{20}$
9	rsi_14	$-\mathrm{RSI}(14)$
10	trend_20	$P_t/\mathrm{MA}_{20} - 1$
11	gap_1	$P_t/P_{t-1} - 1$
12	skew_20	skewness of r_{20}
13	kurt_20	excess kurtosis of r_{20}
14	dvol_20	$-\mathrm{std}(r_{<0})_{20}$
15	maxdd_60	max drawdown over 60 d

All factors are cross-sectionally winsorised (1 %) then z-scored.

Appendix A7: Portfolio Construction & Beta Hedge

Scoring: $\text{score}_i = Z_i^\top w, \quad Z \in \mathbb{R}^{N\times 15}$ (exposures), $w \in \mathbb{R}^{15}$ (PPO action).
Selection: Top-80 by score $\rightarrow$ **long**; Bottom-80 $\rightarrow$ **short**.
Weighting: Within each side, weight proportional to $|\text{score}|$; rescaled so $\sum |h_L| = L_{\text{budget}} = \frac{0.95}{2} K_{\text{ppo}}$, same for short side.
Position bounds: $h_{\min} = 0.5\,\% \cdot K_{\text{ppo}}, \quad h_{\max} = 15\,\% \cdot K_{\text{ppo}}$.

Beta adjustment (up to 40 iterations):

1. Compute $\hat{\beta} = \sum_i h_i \beta_i / K_{\text{ppo}}$.
2. If $|\hat{\beta} - 0| \leq 0.10$, stop.
3. Find best (i, j) pair (one long, one short) whose $\beta_i - \beta_j$ can tilt $\hat{\beta}$ toward target.
4. Shift Δ dollars from j to i subject to position bounds and budget caps.
5. Repeat.

Appendix A8: PPO Reward Function

The agent receives a **PPO-isolated return** (SPY contribution stripped out):

$$r_{\text{ppo}} = \frac{r_{\text{portfolio}} - w_{\text{spy}} \cdot r_{\text{SPY}}}{g_{\text{ppo}}}$$

Composite reward:

$$R = r_{\text{ppo}} - \underbrace{\lambda_\beta \max(0, |\beta| - 0.10)}_{\text{beta penalty}} - \underbrace{\lambda_f \cdot \text{fee\%}}_{\text{commission}} - \underbrace{\lambda_b \cdot \text{borrow\%}}_{\text{borrow cost}} - \underbrace{\eta \cdot \text{turnover\%}}_{\text{turnover}} - \underbrace{\lambda_\tau \cdot \text{tax\%}}_{\text{div WHT}}$$

Coefficient	Backtest	Live
λ_β	2.0	2.0
λ_f	0	1.0
λ_b	0	1.0
η	0	0.05
λ_τ	0	1.0

Backtest trains on pure alpha signal; live adds real-world frictions.

Appendix A9: Transaction Cost Formulas

Slippage (`SpreadSlippageModel`):

$$\text{slip} = P \times 0.00005 \quad \text{per side } (0.5\,\text{bps})$$

Commission (`IBTieredFeeModel`):

$$\text{fee} = \max\Big(\$0.35,\ \ \min(Q \times \$0.0035,\ \ Q \cdot P \cdot 0.005)\Big)$$

Daily borrow charge:

$$\text{borrow}_t = \sum_{i:\, h_i < 0} |h_i \cdot P_i| \ \times\ \frac{0.03}{252}$$

Dividend withholding tax:

$$\text{WHT}_j = D_j \times Q_j \times 0.30 \qquad (Q_j > 0 \text{ only})$$

Total execution cost per period:

$$C = \sum \text{slippage} + \sum \text{fee} + \sum \text{borrow}_t + \sum \text{WHT}_j$$